机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L2 Modeling —— Case Study

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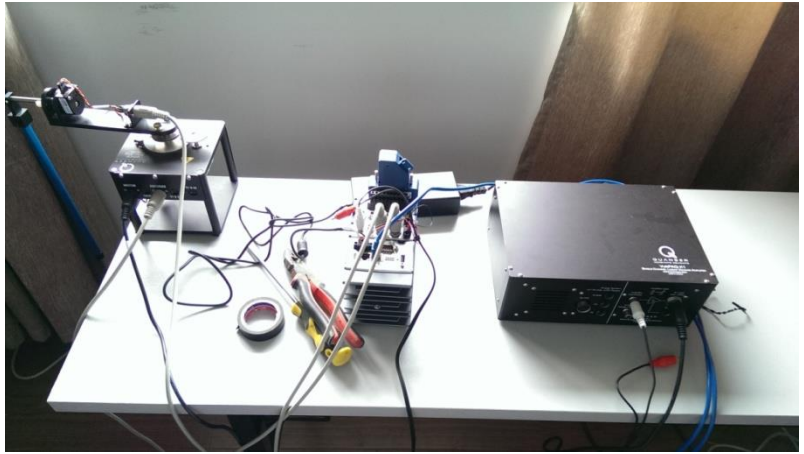
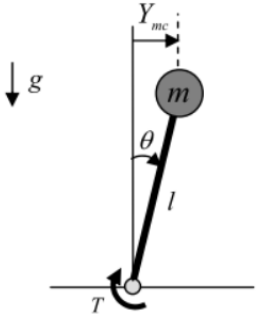
Rotary Inverted Pendulum System

- 倒立摆系统组成
  - 倒立摆
  - 伺服电机
  - 齿轮减速器
  - 电流驱动器
  - CRIO控制器
  - 编码器
- 倒立摆系统应用



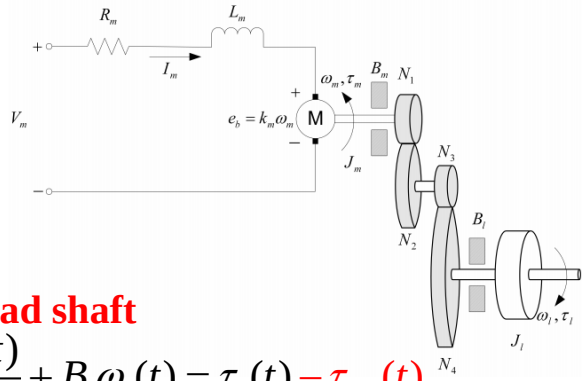


双足机器人一条腿抽象成控制系统中最基本的倒立摆模型

Modeling of Rotary Servo Base Unit

- **Electrical circuit** (电气回路)

$$\begin{cases} L_m \dfrac{dI_m}{dt} = V_m(t) - I_m(t)R_m - k_m\omega_m(t) \\ \tau_m(t) = \eta_m k_t I_m(t) \end{cases} \xrightarrow{\text{忽略}L_m} \tau_m(t) = \frac{\eta_m k_t [V_m(t) - k_m\omega_m(t)]}{R_m}$$



- **Mechanical dynamics** (机械动力学)
  - **Motor shaft**

$$\begin{cases} J_m \dfrac{d\omega_m(t)}{dt} + B_m\omega_m(t) + \tau_{ml}(t) = \tau_m(t) \\ \tau_{ml}(t) = \dfrac{\tau_l(t)}{\eta_g K_g} \end{cases}$$

+

  - **Interconnection law**

$$\begin{cases} K_g = K_{gi}K_{ge} = \dfrac{N_2}{N_1}\dfrac{N_4}{N_3} \\ \theta_m(t) = K_g\theta_l(t) \\ \omega_m(t) = K_g\omega_l(t) \end{cases}$$

+

  - **Load shaft**

$$J_l \frac{d\omega_l(t)}{dt} + B_l\omega_l(t) = \tau_l(t) - \tau_{dis}(t)$$

$$\rightarrow \quad J_r \frac{d\omega_l(t)}{dt} + B_r\omega_l(t) = \eta_g K_g \tau_m(t) - \tau_{dis}(t)$$

$$\begin{cases} J_r = \eta_g K_g^2 J_m + J_l \\ B_r = \eta_g K_g^2 B_m + B_l \end{cases}$$

- **Dynamics of rotary server base** (伺服转台动力学)

$$\boxed{J_t \frac{d^2\theta_l(t)}{dt} + B_t \frac{d\theta_l(t)}{dt} = K_u u(t) - \tau_{dis}(t)}$$

$$\begin{cases} J_t = J_r \\ B_t = \dfrac{\eta_g K_g^2 \eta_m k_t k_m + B_r R_m}{R_m} \\ K_u = \dfrac{\eta_g K_g \eta_m k_t}{R_m} \end{cases}$$

Modeling of Inverted Pendulum

- **Mass center position of the rod** (摆杆质心位置)

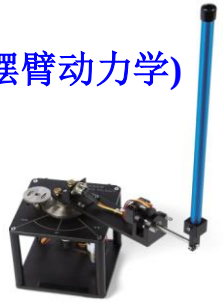
$$\begin{cases} x_p = L_t\theta - L_c\sin\alpha \\ y_p = L_c\cos\alpha \end{cases} \Rightarrow \begin{cases} \dot{x}_p = L_t\dot{\theta} - L_c(\cos\alpha)\dot{\alpha} \\ \dot{y}_p = -L_c(\sin\alpha)\dot{\alpha} \end{cases} \Rightarrow \begin{cases} \ddot{x}_p = L_t\ddot{\theta} - L_c(\cos\alpha)\ddot{\alpha} + L_c(\sin\alpha)\dot{\alpha}^2 \\ \ddot{y}_p = -L_c(\sin\alpha)\ddot{\alpha} - L_c(\cos\alpha)\dot{\alpha}^2 \end{cases}$$

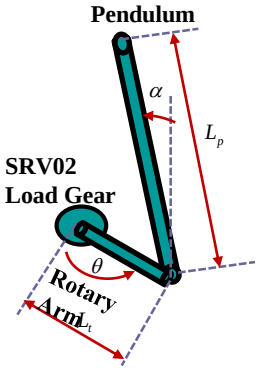
- **Rod Dynamics** (摆杆动力学)

$$\begin{cases} m_p\ddot{x}_p = F_h \\ m_p\ddot{y}_p = F_v - m_p g \\ J_p\ddot{\alpha} = F_v L_c\sin\alpha + F_h L_c\cos\alpha - B_p\dot{\alpha} \end{cases}$$

- **Rotary Arm Dynamics** (摆臂动力学)

$$J_t\ddot{\theta} = K_u u(t) - B_t\dot{\theta} - T_c - F_h L_t$$



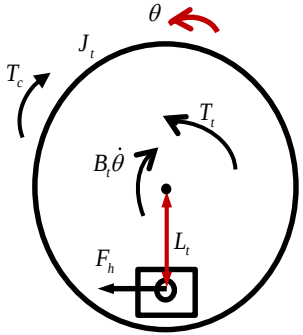


- **Entire Nonlinear dynamics** (非线性动力学)

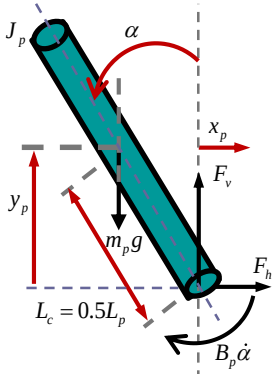
$$\begin{cases} (J_t + m_p L_t^2)\ddot{\theta} = K_u u(t) - T_c - B_t\dot{\theta} + m_p L_t L_c(\cos\alpha)\ddot{\alpha} - m_p L_t L_c(\sin\alpha)\dot{\alpha}^2 \\ (J_p + m_p L_c^2)\ddot{\alpha} = m_p g L_c\sin\alpha + m_p L_t L_c(\cos\alpha)\ddot{\theta} - B_p\dot{\alpha} \end{cases}$$

Assuming α to be small, i.e., $\sin\alpha \approx \alpha, \cos\alpha \approx 1, \dot{\alpha}^2 \approx 0$

$$\begin{bmatrix} J_t + m_p L_t^2 & -m_p L_t L_c \\ -m_p L_t L_c & J_p + m_p L_c^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{bmatrix} = \begin{bmatrix} -B_t\dot{\theta} \\ -B_p\dot{\alpha} + m_p g L_c\alpha \end{bmatrix} + \begin{bmatrix} K_u \\ 0 \end{bmatrix} u(t) - \begin{bmatrix} 1 \\ 0 \end{bmatrix} T_c(t)$$



Rotary arm dynamics



Rod dynamics

Transfer Function

- **Transfer function form *Θ*(s) to *α*(s)** (*Θ*(s) 到 *α*(s)的传递函数)

$$G_r(s)=\frac{\alpha(s)}{\theta(s)}=\frac{K_r\omega_r^{\ 2}s^2}{s^2+2\zeta_r\omega_r s-\omega_r^{\ 2}}$$

$$(J_p+m_pL_c^{\ 2})\ddot{\alpha}+B_p\dot{\alpha}-m_pgL_c\alpha=m_pL_tL_c\ddot{\theta}$$

where, $\zeta_r=\sqrt{\frac{B_p^{\ 2}}{4m_pL_cI_pg}},\omega_r=\sqrt{\frac{m_pL_cg}{I_p}},K_r=\frac{m_pL_cL_t}{I_p\omega_r^{\ 2}}=\frac{L_t}{g},I_p=J_p+m_pL_c^{\ 2}$

- **Transfer function form *u*(s) to *Θ*(s) with disturbance *s*²α(s)** (*Θ*(s) 到 *α*(s)带干扰*s*²α(s)的传递函数)

$$\theta(s)=G_t(s)[u(s)+T_{dis}(s)+\bar{K}_as^2\alpha(s)]$$

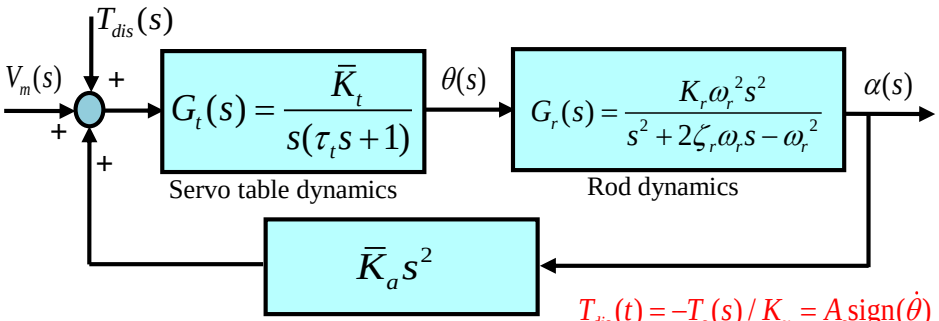
$$(J_t+m_pL_t^{\ 2})\ddot{\theta}=K_uu(t)-T_c(t)-B_t\dot{\theta}+m_pL_cL_t\ddot{\alpha}$$

$$G_t(s)=\frac{\bar{K}_t}{s(\tau_ts+1)},\tau_t=\frac{I_t}{B_t},\bar{K}_t=\frac{K_u}{B_t},\bar{K}_a=\frac{m_pL_tL_c}{K_u},I_t=J_t+m_pL_t^{\ 2},K_u=\frac{\eta_m\eta_gk_tK_g}{R_m}$$

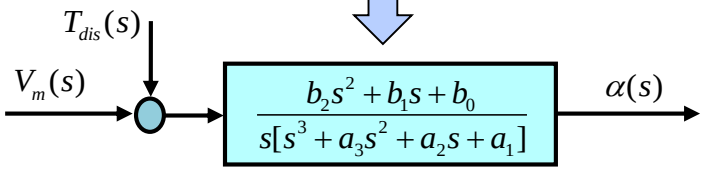
$$T_{dis}(s)=-T_c(s)/K_u$$

Block Diagram

- **Block Diagram**



$$T_{dis}(t) = -T_c(s)/K_u = A_c \operatorname{sign}(\dot{\theta})$$



$$\zeta_r = \sqrt{\frac{B_p^2}{4 m_p L_c I_p g}}, \omega_r = \sqrt{\frac{m_p L_c g}{I_p}}, K_r = \frac{L_t}{g}, \tau_t = \frac{I_t}{B_t}, \bar{K}_t = \frac{K_u}{B_t}, \bar{K}_a = \frac{m_p L_t L_c}{K_u},$$

$$I_t = J_t + m_p L_t^2, I_p = J_p + m_p L_c^2, K_u = \frac{\eta_m \eta_g k_t K_g}{R_m},$$

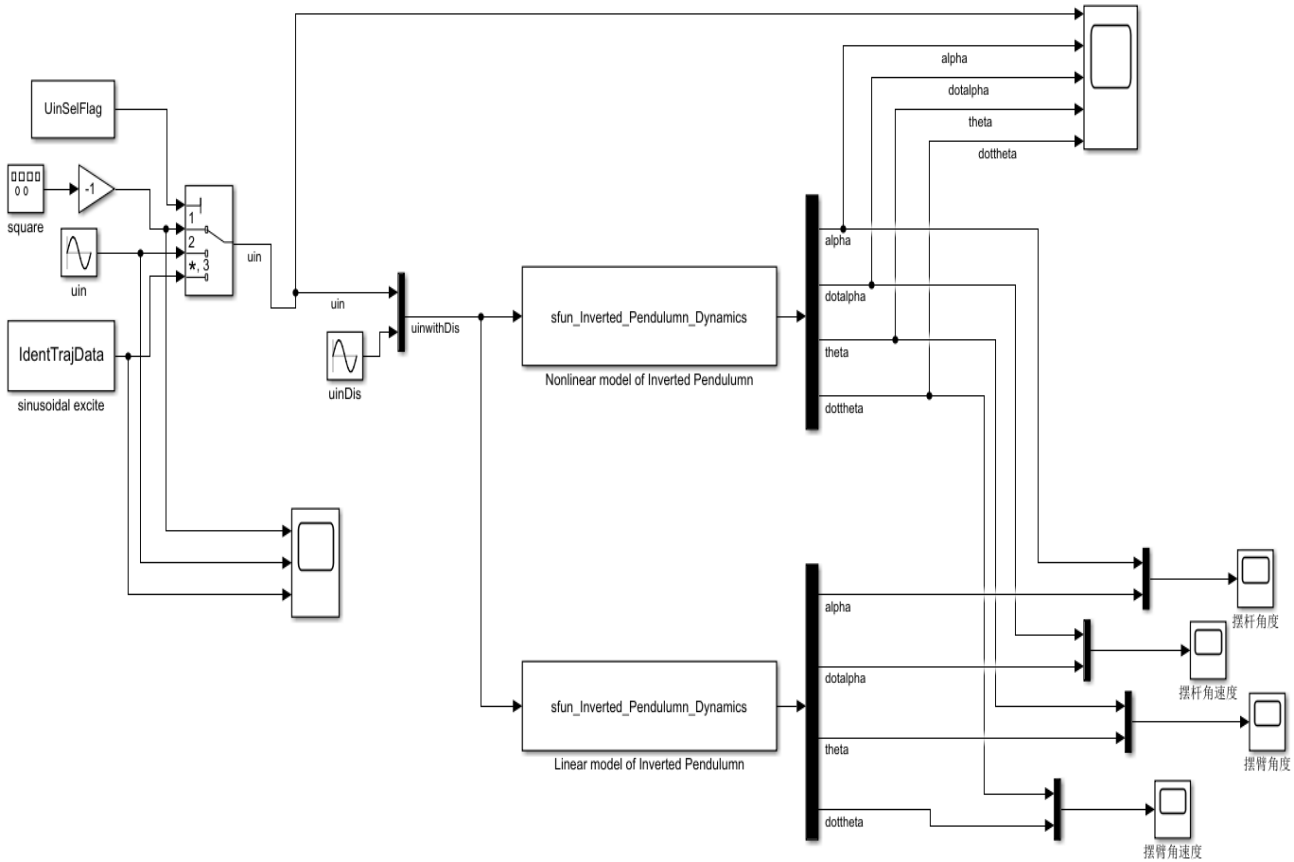
$$\begin{cases} J_t = J_r \\ B_t = \dfrac{\eta_g K_g^2 \eta_m k_t k_m + B_r R_m}{R_m} \end{cases}$$

$$B_t = \frac{\eta_g \eta_m k_t k_m K_g^2 + B_r R_m}{R_m}, J_t = \eta_g K_g^2 J_m + J_l,$$

Physical Parameters of Quanser Rotary Inverted Pendulum

Symbol	Description	Value	Unit
m_p	Mass of pendulum (摆杆质量)	0.127	kg
L_c	Half of the total length of pendulum	0.1685	m
L_p	Total length of pendulum (摆杆长度)	0.337	m
J_p	Pendulum moment of inertia about center of mass (摆杆转动惯量)	0.0012	kg.m^2
B_p	viscous damping coefficient of the pendulum (摆杆阻尼系数)	0.0024	N.m.s/rad
L_t	Rotary arm length from pivot to tip (旋臂长度)	0.216	m
J_r	Rotary arm moment of inertia about its center of mass (旋臂等效转动惯量)	0.001	kg.m^2
B_r	Rotary arm viscous damping coefficient as seen at the pivot axis (旋臂等效阻尼系数)	0.0024	N.m.s/rad
R_m	Motor armature resistance (电机电阻)	2.6	Ω
k_t	Motor current-torque constant (电流转矩系数)	7.68 ×10^{-3}	N.m/A
k_m	Motor back-EMF constant (反电动势系数)	7.68×10^{-3}	V/(rad/s)
K_g	Total gear ratio (减速器传动比)	70	
η_m	Motor efficiency (电机效率)	1	
η_g	Gearbox efficiency (减速器效率)	1	
V_m	Applied voltage to the motor (电机输入电压)	-10~10	V
T_c	Coulomb friction torque of the bearings for the table rotation and the motor shaft (摩擦力矩)		N.m
A_c	Coefficient of Coulomb friction torque	0.4	V

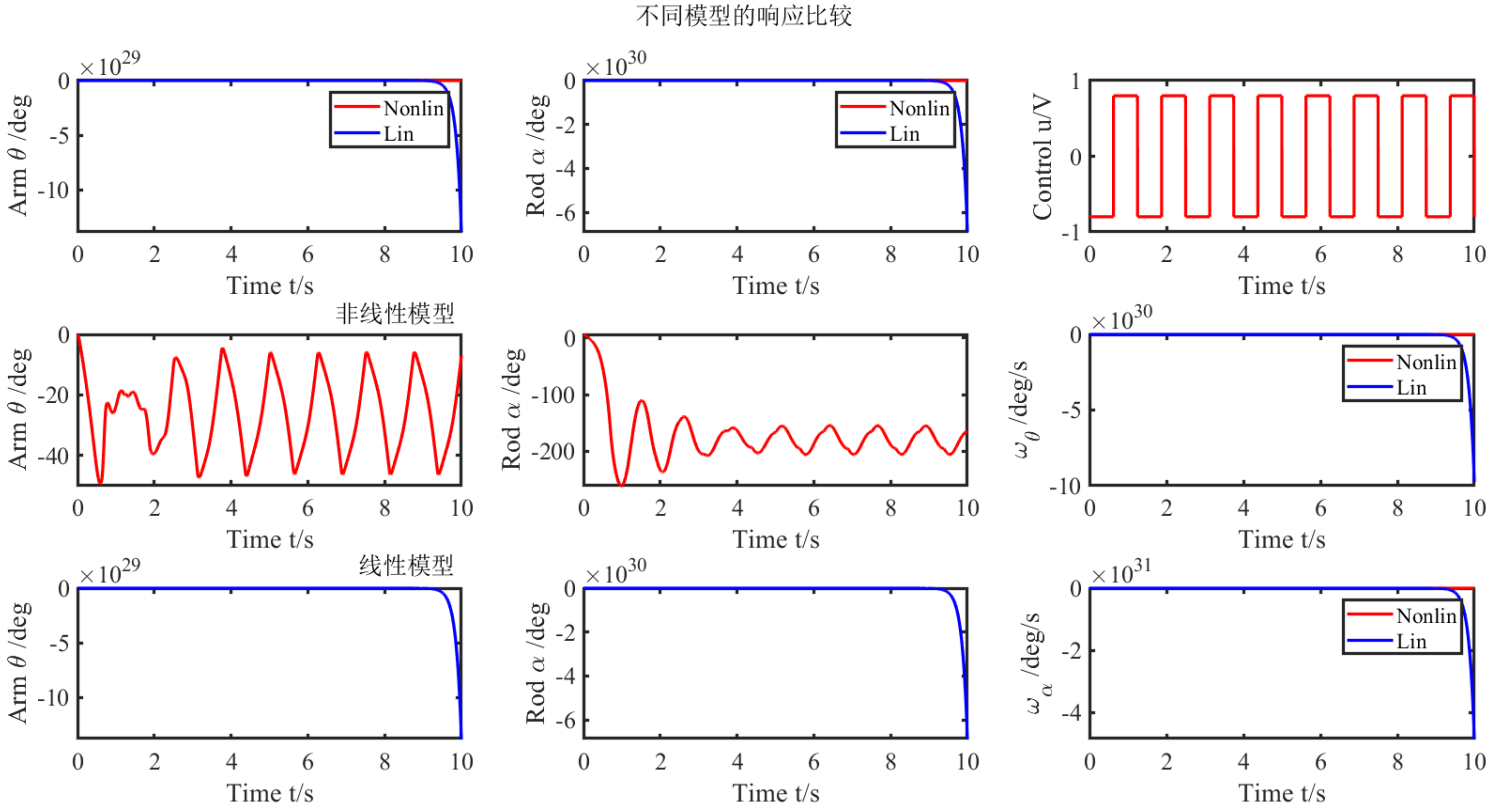
Comparison of Nonlinear and Linear Models




```
signcoef = 1e3;
MM3 = [ 0
        Ku*Voltage];
MM4 = [0
       -uinDis-Ku*AcFric*tanh(theta_dot*signcoef)];
if(ModelSelFlag == 0) %Sero table without rod
   sys(1,1) = 0;
   sys(2,1) = 0;
   sys(3,1) = theta_dot;
   sys(4,1) = (Ku*Voltage-uinDis-Ku*AcFric*tanh(theta_dot*signcoef)-Bt*theta_dot)/Jt;
   return;
elseif(ModelSelFlag == 1) %nonlinear Inverted Pendulum
   MM1 = [Ip        -A3*cos(alpha)
       -A3*cos(alpha)     It        ];
   MM2 = [-Bp*alpha_dot + A2*sin(alpha)
        -Bt*theta_dot - A3*sin(alpha)*alpha_dot^2];
elseif(ModelSelFlag == 2) %linear Inverted Pendulum
   MM1 = [Ip     -A3
        -A3      It   ];
   MM2 = [-Bp*alpha_dot + A2*alpha
        -Bt*theta_dot];
 elseif(ModelSelFlag == 3) %nonlinear pendulum
    MM1 = [Ip              A3*cos(alpha)
           A3*cos(alpha)      It         ];
    MM2 = [-Bp*alpha_dot - A2*sin(alpha)
           -Bt*theta_dot - A3*sin(alpha)*alpha_dot^2];
 elseif(ModelSelFlag == 4) %linear pendulum
    MM1 = [Ip     A3
           A3     It   ];
    MM2 = [-Bp*alpha_dot - A2*alpha
           -Bt*theta_dot];
 else
     MM1=diag([1,1]');
     MM2 = [0,0]';
     MM3 = [0,0]';
     MM4 = [0,0]';
 end
 yy = inv(MM1)*(MM2 + MM3 + MM4);
 dx1 = alpha_dot;
 dx2 = yy(1);
 dx3 = theta_dot;
 dx4 = yy(2);
```

Comparison of Nonlinear and Linear Models

- **Pendulum system, initial α being 6° with square exciting signal of amplitude being 0.8V**



Modeling of Pendulum (Rod downward)

- **Mass center position of the rod (rod downward)**

$$\begin{cases} x_{p1} = L_t\theta + L_c \sin\alpha_1 \\ y_{p1} = -L_c \cos\alpha_1 \end{cases} \Rightarrow \begin{cases} \dot{x}_p = L_t\dot{\theta} + L_c(\cos\alpha_1)\dot{\alpha}_1 \\ \dot{y}_p = L_c(\sin\alpha_1)\dot{\alpha}_1 \end{cases} \Rightarrow \begin{cases} \ddot{x}_{p1} = L_t\ddot{\theta} + L_c(\cos\alpha_1)\ddot{\alpha}_1 - L_c(\sin\alpha_1)\dot{\alpha}_1^{\,2} \\ \ddot{y}_{p1} = L_c(\sin\alpha_1)\ddot{\alpha}_1 + L_c(\cos\alpha_1)\dot{\alpha}_1^{\,2} \end{cases}$$

- **Rod Dynamics**

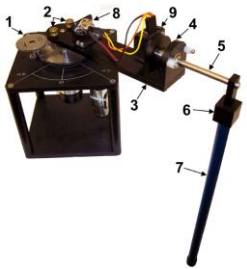
$$\begin{cases} m_p\ddot{x}_{p1} = F_h \\ m_p\ddot{y}_{p1} = F_v - m_p g \\ J_p\ddot{\alpha}_1 = -F_v L_c \sin\alpha_1 - F_h L_c \cos\alpha_1 - B_p\dot{\alpha}_1 \end{cases}$$

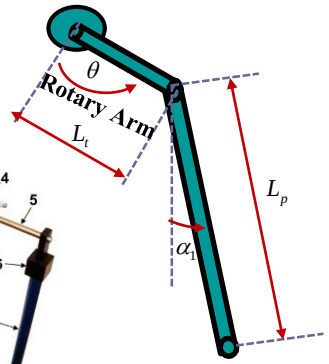
$\boxed{+}$

- **Rotary Arm Dynamics**

$$J_t\ddot{\theta} = K_u u - B_t\dot{\theta} - T_c - F_h L_t$$

$$F_h = m_p L_t\ddot{\theta} + m_p L_c\ddot{\alpha}$$



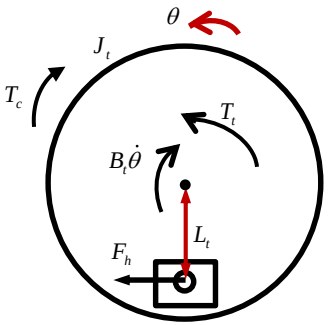


- **Entire Nonlinear dynamics**

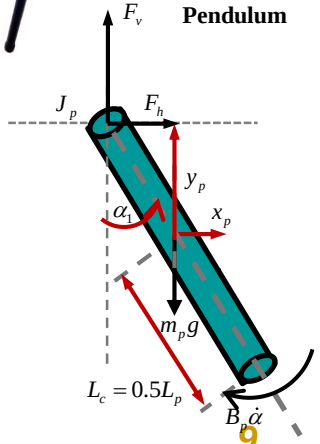
$$\Rightarrow \begin{cases} (J_t + m_p L_t^{\,2})\ddot{\theta} = K_u u - T_c - B_t\dot{\theta} - m_p L_t L_c(\cos\alpha_1)\ddot{\alpha}_1 - m_p L_t L_c(\sin\alpha_1)\dot{\alpha}_1^{\,2} \\ (J_p + m_p L_c^{\,2})\ddot{\alpha}_1 = -m_p g L_c \sin\alpha_1 - m_p L_t L_c(\cos\alpha_1)\ddot{\theta} - B_p\dot{\alpha}_1 \end{cases}$$

Assuming α_1 to be small, i.e., $\sin\alpha_1 \approx \alpha_1, \cos\alpha_1 \approx 1, \dot{\alpha}_1^{\,2} \approx 0$

$$\boxed{\begin{bmatrix} J_t + m_p L_t^{\,2} & m_p L_t L_c \\ m_p L_t L_c & J_p + m_p L_c^{\,2} \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha}_1 \end{bmatrix} = \begin{bmatrix} -B_t\dot{\theta} \\ -B_p\dot{\alpha}_1 - m_p g L_c \alpha_1 \end{bmatrix} + \begin{bmatrix} K_u \\ 0 \end{bmatrix} u(t) - \begin{bmatrix} 1 \\ 0 \end{bmatrix} T_c(t)}$$



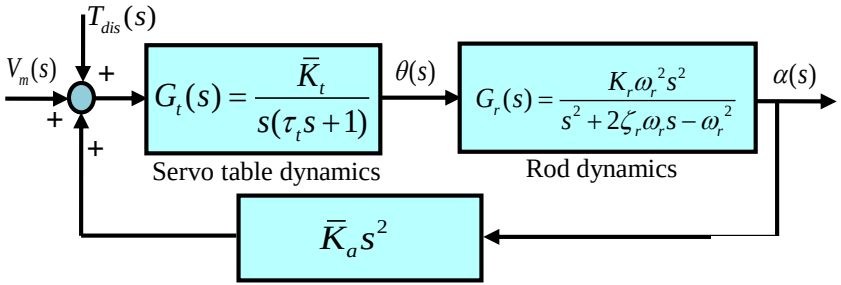
Rotary arm dynamics



Rod dynamics

Comparison of Transfer Function

- **Rod upward (倒摆)**



$$G_r(s)=\frac{\alpha(s)}{\theta(s)}=\frac{K_r\omega_r^2 s^2}{s^2+2\zeta_r\omega_r s-\omega_r^2}$$

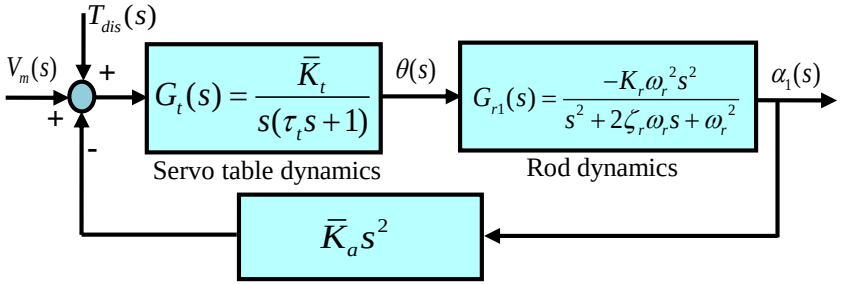
$$\theta(s)=G_t(s)[u(s)+T_{dis}(s)+\bar{K}_a s^2\alpha(s)]$$

- **Entire nonlinear dynamics**

$$\begin{bmatrix} J_t+m_pL_t^2 & -m_pL_tL_c(\cos\alpha) \\ -m_pL_tL_c(\cos\alpha) & J_p+m_pL_c^2 \end{bmatrix}\begin{bmatrix}\ddot{\theta}\\ \ddot{\alpha}\end{bmatrix}$$
$$=\begin{bmatrix}-B_t\dot{\theta}-m_pL_tL_c(\sin\alpha)\dot{\alpha}^2\\ -B_p\dot{\alpha}+m_pgL_c\sin\alpha\end{bmatrix}+\begin{bmatrix}K_u\\0\end{bmatrix}u(t)-\begin{bmatrix}K_u\\0\end{bmatrix}T_{dis}(t)$$

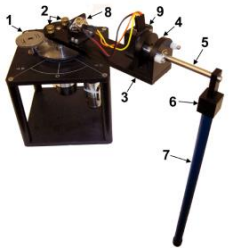
$$\alpha_1=\alpha-\pi$$

- **Rod downward (单摆)**



$$G_{r1}(s)=\frac{\alpha_1(s)}{\theta(s)}=\frac{-K_r\omega_r^2 s^2}{s^2+2\zeta_r\omega_r s+\omega_r^2}$$

$$\theta(s)=G_t(s)[u(s)+T_{dis}(s)-\bar{K}_a s^2\alpha_1(s)]$$

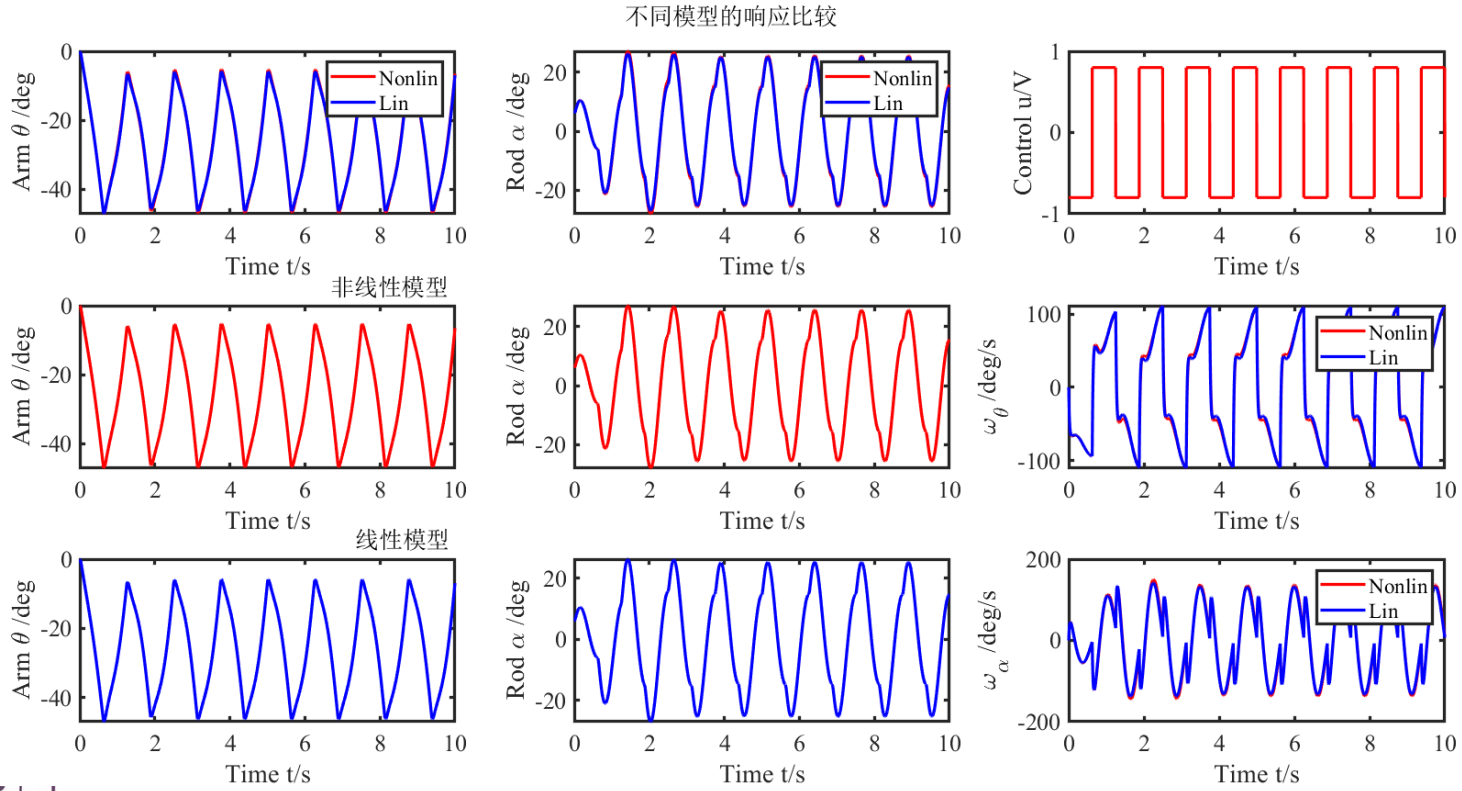


- **Entire nonlinear dynamics**

$$\begin{bmatrix} J_t+m_pL_t^2 & m_pL_tL_c(\cos\alpha_1) \\ m_pL_tL_c(\cos\alpha_1) & J_p+m_pL_c^2 \end{bmatrix}\begin{bmatrix}\ddot{\theta}\\ \ddot{\alpha}_1\end{bmatrix}$$
$$=\begin{bmatrix}-B_t\dot{\theta}-m_pL_tL_c(\sin\alpha)\dot{\alpha}_1^2\\ -B_p\dot{\alpha}_1-m_pgL_c\sin\alpha_1\end{bmatrix}+\begin{bmatrix}K_u\\0\end{bmatrix}u(t)-\begin{bmatrix}K_u\\0\end{bmatrix}T_{dis}(t)$$

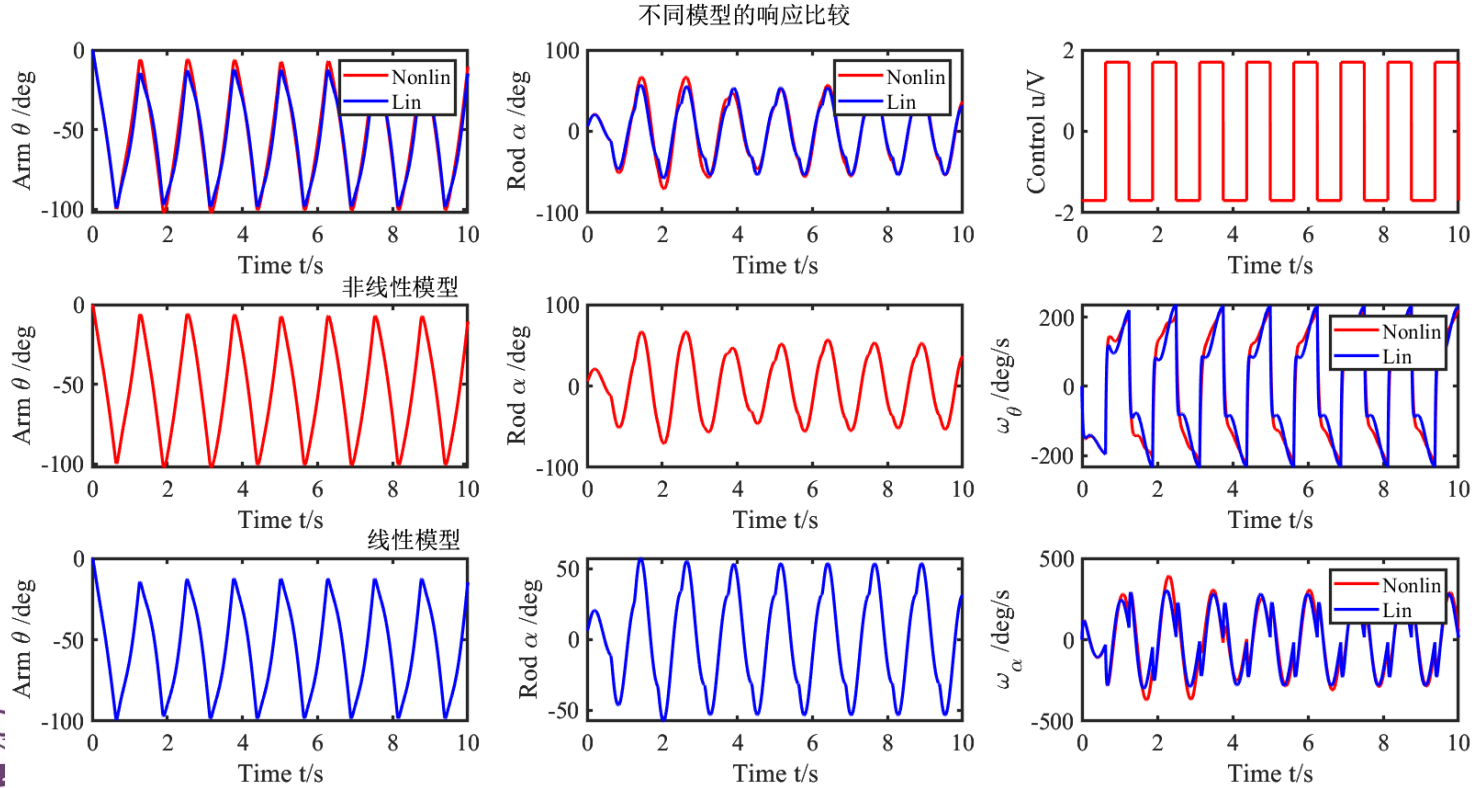
Comparison of Nonlinear and Linear Models

- **Pendulum system, initial α being 6° with square exciting signal of amplitude being 0.8V**



Comparison of Nonlinear and Linear Models

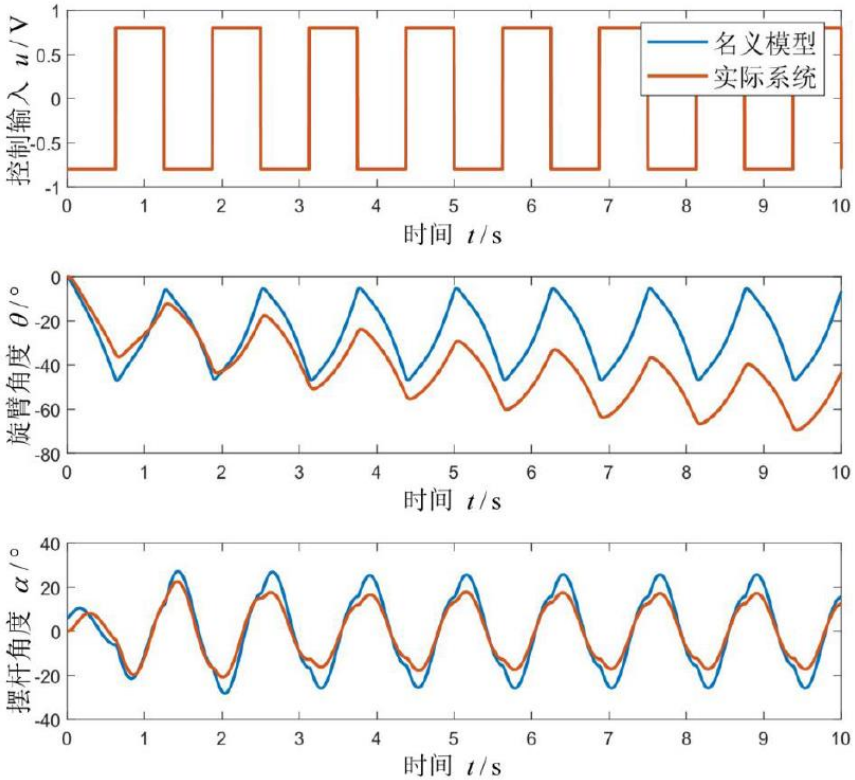
- **Pendulum system, initial α being 6° with square exciting signal of amplitude being 1.7V**



The nonlinear and linear models are more consistent with small exciting signals due to small rod angles response!

Comparison of Experiment and Simulation

➢ **Response of Pendulum system in simulation and experiment**



What results in the difference between the simulation and experiment?